What tumour gene expression reveals about the clinical receptor groups of breast cancer: a TCGA analysis checked in protein and an independent cohort

BT3041 Analysis and Interpretation of Biological Data · Term project report
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Team 16: Vijayavallabh · Vedant · Harshit · Navya · Anitha

Abstract. Breast cancers are assigned to treatment groups by three clinical tests: oestrogen receptor (ER), progesterone receptor (PR) and HER2. We asked what tumour RNA-seq tells us about these clinical groups. We used the clinical labels as ground truth rather than the PAM50 subtypes, because PAM50 is itself derived from expression. In 1,088 TCGA tumours (HR+/HER2− 591, HER2+ 160, triple-negative 157), clustering without labels found a basal-like group that contained 138 of the 157 triple-negative tumours (ARI with ER status 0.67). Compared with HR+/HER2− tumours, triple-negative tumours differed more than 2-fold in 2,741 genes (FDR 5%). The oestrogen-response programme was lower and the proliferation, MYC and immune programmes were higher, but only the oestrogen-response shift remained significant in a correlation-aware gene-set test. Classifiers trained on the whole transcriptome performed as well as the receptor gene alone, not better (ER: *ESR1* AUC 0.959 vs best model 0.954; HER2: *ERBB2* 0.845 vs 0.851). The results agreed with protein levels in the same tumours (mRNA–protein effect correlation ρ = 0.80) and with 1,980 METABRIC tumours, where the frozen models reached AUC 0.96 for ER and 0.91 for HER2 and 86% of the differentially expressed genes replicated.

1. Introduction and research questions

Pathologists test every invasive breast cancer for ER and PR by immunohistochemistry (IHC) and for HER2 by IHC or in-situ hybridisation (FISH) [1, 2]. These results decide the treatment: endocrine drugs for hormone-receptor-positive (HR+) disease, anti-HER2 antibodies for HER2+ disease, and chemotherapy for triple-negative breast cancer (TNBC), for which there is no targeted option. Expression profiling defined the intrinsic subtypes (luminal A/B, HER2-enriched, basal-like) [3, 4], which are now assigned with the PAM50 centroids [5]. Many machine-learning studies predict PAM50 from RNA-seq. Since PAM50 is computed from RNA in the first place, part of their accuracy simply reflects an algorithm agreeing with itself. For this reason we treated the clinical IHC/FISH labels as ground truth and used PAM50 only for comparison.

We set out to answer three questions. **Q1:** Does the structure of tumour expression, found without labels (PCA, UMAP, clustering), recover the clinical receptor groups? **Q2:** Which genes and pathways differ between TNBC and HR+/HER2− tumours, and how robust are these differences? **Q3:** Can whole-transcriptome classifiers predict clinical ER and HER2 status better than the receptor gene on its own? We checked each answer in three ways: in a second assay (RPPA protein) on the same tumours, in an independent cohort (METABRIC) whose labels come from other assays, and against published numbers.

2. Data acquisition and preprocessing

Sources. TCGA-BRCA RNA-seq (HiSeqV2, log2(RSEM+1)), STAR gene counts (GDC, GENCODE v36), clinical IHC/FISH fields, reverse-phase protein arrays (RPPA) and PAM50 calls were downloaded from UCSC Xena [6]; survival end points came from the TCGA Clinical Data Resource [7]. METABRIC Illumina HT-12 expression and clinical tables [8, 9] were taken from the cBioPortal datahub. Gene symbols in all data sets, including the Hallmark gene sets [10], were mapped to current HGNC symbols, and the raw files were checksummed at a data freeze.

Cleaning and labels. We kept one primary tumour per patient (1,097) and excluded 9. Five of these had HiSeqV2 and STAR profiles that did not match (Spearman ρ below the 99th percentile of between-patient correlations), and five have a redacted clinical record in the Clinical Data Resource; one tumour failed both checks. This left 1,088 tumours for all label-free work. HER2 status followed ASCO/CAP 2018 [2]. FISH was used first, then the IHC score (3+ positive, 0/1+ negative, 2+ without FISH missing), and the status text only when no score was recorded. Tumours that were IHC 3+ but FISH-negative (5) were treated as discordant and left unlabelled, as were other conflicting records. This rule agrees with the TCGA 2012 HER2 calls at κ = 0.88. The supervised set contains the 1,076 female patients; the 12 male patients were used only in label-free work. The groups were HR+/HER2− (ER or PR positive, HER2 negative), HER2+ (any

ER/PR) and TNBC (all three negative), and 168 tumours were missing one of the three results. For differential expression we kept the 24,660 STAR genes with ≥ 10 counts in at least 157 tumours (the size of the smallest group). We did not scale the matrices at this stage, so that every scaling step could be done inside the cross-validation folds. Table 1 lists the data sets.

Table 1. Data sets and group sizes under the final labels (HER2 rule v1). *METABRIC groups use IHC ER and SNP6 copy-number HER2 (gain vs no gain); PR is not available, so the ER−/HER2- not-gained group is not called TNBC.

Data set	Used for	n	HR+/ HER2−	HER2+	TNBC
TCGA HiSeqV2 (analysed)	PCA, UMAP, clustering	1,088	596	166	157
TCGA supervised set	classifiers	1,076	591	160	157
TCGA STAR counts	differential expression	1,075	591	160	157
TCGA RPPA (131 antibodies)	protein validation	730	395	114	109
METABRIC (HT-12 arrays)	external validation	1,980	1,220*	426*	286*

3. Methods

All analyses were run in Python 3.12 (scikit-learn, PyDESeq2, GSEApy, lifelines) with seed 42. Each result folder has a params.json that records every parameter and package version, and the code is in the supplementary material. P-values from multiple tests were corrected with the Benjamini–Hochberg (BH) FDR [11].

Exploration and clustering (Q1). PCA and UMAP [12] (15 neighbours) were computed on the 2,000 most variable genes after sex-chromosome-specific genes were removed. Marker genes were compared between groups with Mann–Whitney U tests, using rank-biserial r as the effect size. We also measured how much PC variance was explained by group and by tissue-source site (η^2, against a permutation chance level). k-means (k = 2–8, without the PAM50 genes) was combined with consensus clustering [13] over 100 resamples of 80% of the tumours. We fixed k before looking at the labels, as the value with the lowest proportion of ambiguous clustering (PAC) [14]. Agreement with the labels was measured by the adjusted Rand index (ARI) with bootstrap 95% CIs, and compared with the best ARI that two clusters can reach.

Differential expression and pathways (Q2). We ran PyDESeq2 [15, 16] on raw STAR counts with the design ~ histology + group. It tested |log2FC| > 1 directly (lfc_null = 1), instead of testing for zero change and filtering by fold change afterwards. Shrunk fold changes were used for ranking and plotting. In sensitivity runs we added age, kept only invasive ductal tumours, dropped every low-confidence HER2 call, or used the ER−/HER2− vs ER+/HER2− contrast that matches METABRIC. For Hallmark enrichment we used GSEA prerank on the Wald statistic [17, 18] (1,000 permutations) and Fisher over-representation of the DEG lists. Two further tests take inter-gene correlation into account: CAMERA [19] and phenotype-permutation GSEA.

Classification (Q3). Targets were clinical ER (n = 1,024, 789 positive) and HER2 (n = 913, 161 positive). We trained L1-penalised logistic regression, a linear SVM, a Random Forest and PLS-DA in nested cross-validation (outer 5 folds × 3 repeats, inner 5 folds, class-balanced weights). The variance filter (top 5,000 genes), scaling, univariate selection of k genes ($k \in \{50, 200, 1000\}$) and the hyper-parameters were all fitted inside the training folds. The baseline was the receptor gene on its own (*ESR1* or *ERBB2*), with its cut-off also chosen inside the folds. We compared models on pooled out-of-fold ROC-AUC, using the Nadeau–Bengio corrected t-test [20] and a paired bootstrap. An ablation removed *ESR1*, *PGR*, *ERBB2*, every gene within 1.5 Mb (and 2 Mb) of *ERBB2* and the 50 PAM50 genes (135 genes in all). Models trained on shuffled labels served as a negative control.

Validation. (i) Protein. We ran Mann–Whitney tests on the RPPA data. For the 85 genes with a total-protein antibody, we correlated DE fold changes with protein differences (Spearman) and compared the result with a gene-label permutation null. We also compared the out-of-fold ER

errors with ERα protein levels. (ii) METABRIC. The ER, HER2 and ER−/HER2− models were refit on all TCGA tumours and frozen, with checksums logged, before any METABRIC label was read. They were then applied to METABRIC after z-scoring within the cohort on the 18,936 shared genes. DE was replicated by OLS with age as a covariate (ER-IHC−/HER2-not-gained 286 vs ER-IHC+/HER2-not-gained 1,220). (iii) Survival. We used Kaplan–Meier curves, log-rank tests and Cox models with Schoenfeld checks. The TCGA model of progression-free interval (PFI) included group, age and stage (100 events, which allows 5 parameters); the METABRIC model of breast-cancer death within 10 years included group, age, grade and size. (iv) Literature. We placed our numbers next to published ones and checked each source against PubMed.

4. Results and interpretation

4.1 Label-free structure follows ER, and recovers TNBC as a basal-like group (Q1)

PC1 (14% of variance) was the ER axis. It separated IHC ER-positive from ER-negative tumours with AUC 0.95, and the clinical group explained $\eta^2 = 0.59$ of its variance. Tissue-source site, by contrast, explained ≤ 0.08 of PC1–4. In the UMAP (Fig. 1a), a separate island of 189 tumours contained 129 of the 157 TNBC (82%). HER2+ tumours did not form an island of their own; they sat at one edge of the HR+ mass. The marker genes behaved as expected. *ESR1* showed the largest single difference between TNBC and HR+/HER2− (rank-biserial r = −0.95), the basal transcription factor *FOXC1* was higher in TNBC (r = +0.83), and *ERBB2* was higher in HER2+ (r = +0.65; all $q < 10^{-30}$).

Consensus clustering was most stable at k = 2 (PAC 0.02, compared with 0.14 at k = 5). The smaller of the two clusters had 217 tumours and contained 138 of the 157 TNBC (Fig. 1b). Its agreement with IHC ER was ARI 0.67 (0.61–0.73), and with the three clinical groups 0.42, where the ceiling for two clusters is 0.88 (Fig. 1c). k-means, consensus, Ward and average-linkage clustering gave the same picture at k = 2 (ARI 0.82–1.00). A HER2-enriched cluster appeared at k = 5, but it was the least reproducible across methods. The luminal tumours also split differently from PAM50 LumA/LumB (ARI 0.35), which fits the luminal continuum described by Netanely et al. [21]. **Answer to Q1:** expression recovers the ER-negative, basal-like axis strongly, but the HER2+ group only weakly.

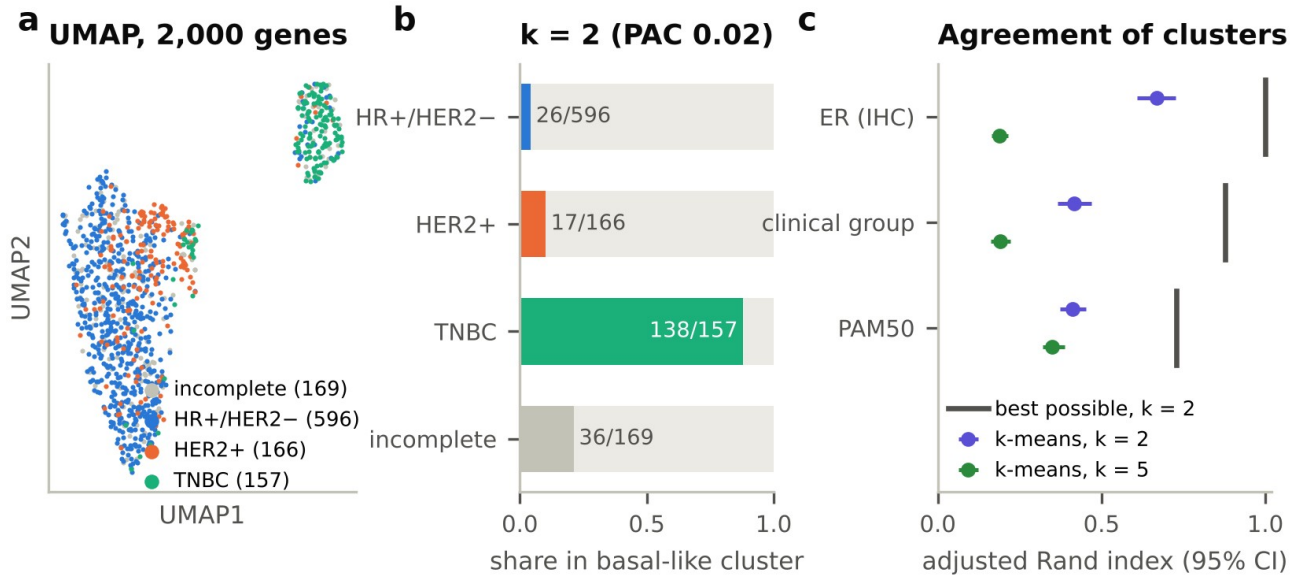


Figure 1. Label-free structure. (a) UMAP of 1,088 tumours coloured by clinical group. (b) Share of each clinical group in the smaller k-means cluster (k = 2). (c) Adjusted Rand index of k-means clusters with ER, the clinical groups and PAM50 (95% bootstrap CI); bars mark the best ARI reachable with two clusters.

4.2 TNBC is a distinct transcriptional state (Q2)

Of 23,439 genes, 2,741 differed more than 2-fold between TNBC and HR+/HER2− at FDR 5% (1,484 higher, 1,257 lower in TNBC; Fig. 2a). A plain Wald test of zero change would have called 18,604 genes (79%); this is why we tested the threshold instead of applying it afterwards. *ESR1* (log2FC −5.1) and *PGR* (−4.4) served as positive controls. Beyond these two, the ER-driven luminal programme was lower in TNBC (*GATA3* −3.3, *AGR3* −4.8, *FOXA1* −2.9, *XBP1* −2.4) and

basal transcription factors and keratins were higher (*FOXC1* +3.1, as reported by Ray et al. [22]; *KRT5* +2.4). The list held up in the sensitivity runs. Of the DEGs, 94% stayed significant with the same sign after adding age, 91% in ductal tumours only (so lobular biology is not being reported as HR+ biology), 87% with strict labels and 96% in the ER−/HER2− contrast. HER2+ and HR+/HER2− tumours differed in only 148 genes, and 33 of these lie within 2 Mb of *ERBB2* on the 17q12 amplicon.

GSEA found 28 Hallmark sets at FDR < 0.05 (Fig. 2b). Proliferation (E2F targets, NES +3.11; G2M checkpoint +2.88), MYC targets (+2.74), interferon-γ (+2.18) and mTORC1 (+2.16) were higher in TNBC. The early and late oestrogen responses were lower (−3.08, −2.40), as in the TCGA and CPTAC proteogenomic studies [4, 23]. Gene-permutation q-values are too optimistic when the genes in a set are co-expressed, and here the median inter-gene correlation was 0.06 (up to 0.25 for E2F targets). Both correlation-aware tests gave the same direction as prerank for all 28 sets (NES ρ = 0.92), but only the early oestrogen response remained significant (CAMERA q < 0.05). We therefore describe the proliferation, MYC and immune programmes as consistent shifts with large effect sizes. We do not claim that they are shifted more than the average gene. **Answer to Q2:** in TNBC the oestrogen-response programme is turned down and basal and proliferative genes are turned up. HER2+ tumours differ much less, mainly at the amplicon.

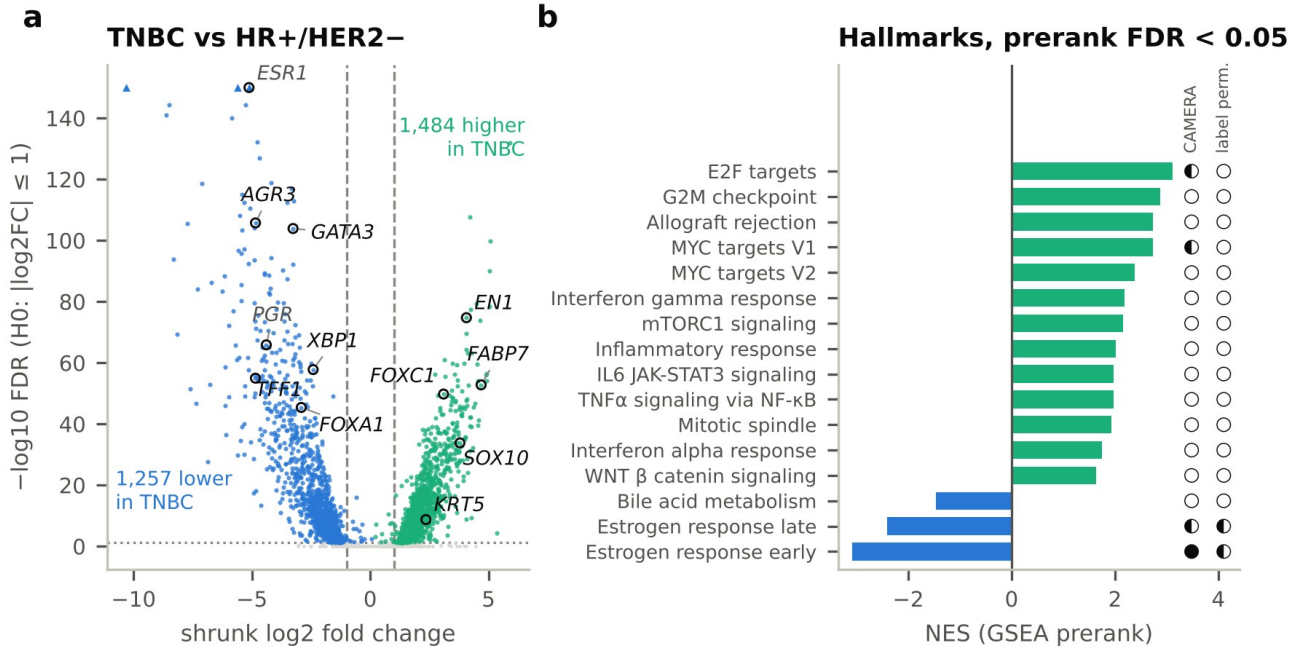


Figure 2. TNBC vs HR+/HER2− (157 vs 591). (a) Volcano plot: shrunk log2 fold change against FDR of the |log2FC| > 1 test (y capped at 150). (b) Hallmark sets at prerank FDR < 0.05 (the 13 strongest positive and all 3 negative of 28); symbols give the correlation-aware checks (● q < 0.05, ◖ nominal p < 0.05, ○ neither).

4.3 Whole-transcriptome models match, but do not beat, the receptor gene (Q3)

For ER, *ESR1* alone reached AUC 0.959 and the best model (Random Forest) 0.954 (corrected t-test p = 0.69). For HER2, *ERBB2* alone reached 0.845 and the best model 0.851 (p = 0.66) (Fig. 3a–b). The four model types gave similar AUCs (Fig. 3c), and shuffled labels gave MCC −0.01 (ER) and +0.04 (HER2). The single-gene accuracy (94% for *ESR1*, 89% for *ERBB2*) matched the 88–96% that Gong et al. reported for arrays [24]. The ablation treated the two receptors very differently. With the receptor genes, the amplicon and the PAM50 genes removed, ER was predicted as well as before (AUC 0.948–0.953), so the ER signal is spread across a whole regulon rather than carried by one gene. HER2 dropped to 0.784–0.820, and the 2 Mb ablation gave the same result. Most of the HER2 signal therefore sits in the amplicon, and what remains largely reflects the lower ER expression and higher proliferation of HER2+ tumours. Label quality also limits HER2. When we dropped the 30 HER2+ calls that rest on status text alone, *ERBB2* alone rose to 0.894 and the best model to 0.884. The combined ER−/HER2− target was the only one where a model beat a single gene (0.964 vs 0.948 for *ESR1*; p = 0.005), since no single gene encodes both receptors. As a secondary (and circular) benchmark, the same pipeline reproduced

the 5-class PAM50 calls with 88% accuracy (87% without the PAM50 genes). **Answer to Q3:** no. The receptor gene on its own already carries almost all the information about clinical status.

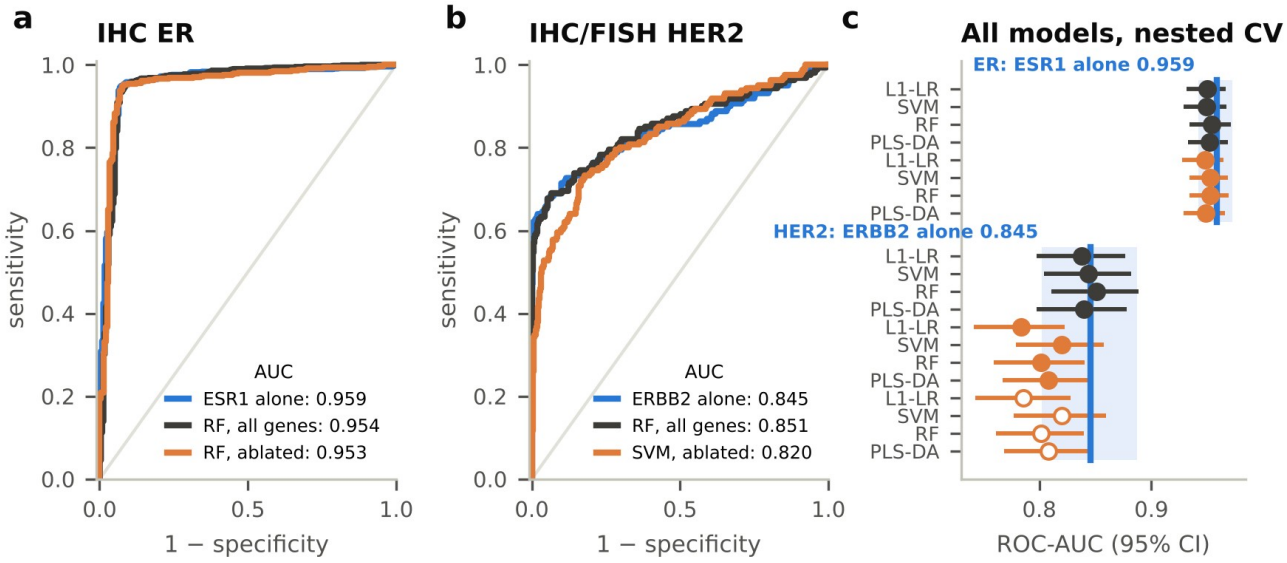


Figure 3. Classification of clinical status (nested CV, pooled out-of-fold). (a, b) ROC curves for the receptor gene alone, the best model with all genes and the best ablated model. (c) AUC (95% CI) of every model: dark, all genes; orange, ablated (±1.5 Mb); open, ablated (±2 Mb). Blue line and band: receptor gene alone.

4.4 Validation in protein, in METABRIC and against published studies

Protein. Of the 131 RPPA antibodies, 92 differed between TNBC and HR+/HER2− ($q < 0.05$). For the 85 genes with a total-protein antibody, the mRNA fold change predicted the protein difference with $\rho = 0.80$ (permutation $p = 0.001$; 95th percentile of the null 0.49; Fig. 4a), and all 15 DEGs with an antibody agreed in sign. The ER errors are informative (Fig. 4b). 15 IHC ER-negative tumours were called ER-positive, and 8 of 10 of them with RPPA had ERα protein above the 95th percentile of correctly called ER-negative tumours. This points to IHC false negatives rather than model errors. Of the 68 IHC ER-positive tumours called negative, 40 were PAM50 Basal or HER2-enriched, and 26 of 43 with RPPA had ERα below the 5th percentile of correctly called ER-positive tumours. These fit ER-low tumours or IHC false positives.

METABRIC. The frozen TCGA models lost no performance in METABRIC (Fig. 4c). ER reached AUC 0.959 (0.947–0.970) against 0.950 in TCGA cross-validation, HER2 (copy-number gain) 0.906 against 0.838, and ER−/HER2− vs rest 0.935. HER2 did better in METABRIC than in cross-validation, most likely because a copy-number label follows mRNA more closely than clinical IHC/FISH does. As in TCGA, the models did not beat the single gene (*ESR1* 0.960, *ERBB2* 0.900). Of the 1,857 TCGA DEGs measured on the array, 1,599 (86%) replicated with the same sign at $q < 0.05$ (effect-size $\rho = 0.79$), and the Hallmark NES values agreed with $\rho = 0.94$.

Table 2. Our numbers next to published ones (each checked against PubMed). TCGA 2012 analysed a subset of the same tumours (a consistency check); the other studies used different cohorts or assays.

Quantity	This study	Published
TNBC called PAM50 Basal	83% (130/157)	86% (65/76) [4]
PAM50 Basal that are TNBC	81% (130/160)	80% (65/81) [4]
Clinical HER2+ called HER2-enriched	33%	~50% [4]
ER+ called Basal or HER2-enriched	6.3%	8.7% [25]
Single-gene accuracy, ER / HER2	94% / 89%	88–96% / 89–93% [24]
AUC change across cohorts, ER / HER2	0.95 → 0.96 / 0.84 → 0.91	13–22% drop [26]
5-y BC death HR vs LumA (ER+, endocrine-treated): LumB / HER2E / Basal	2.6 / 3.4 / 8.2	2.0 / 3.6 / 17.7 [25]

Published studies. The clinical–PAM50 concordance, the single-gene accuracy and the order of risk across subtypes all agree with the literature (Table 2). Clinical HER2+ tumours were called HER2-enriched less often than in TCGA 2012. With the 2012 HER2 labels and PAM50 calls, the same tumours give 52%, so the gap comes from the larger cohort, the newer PAM50 calls and the HER2 rule rather than from an error. Shah and Patel reported a 13–22% loss when a multimodal model was moved from TCGA to CPTAC [26]; our simpler mRNA models lost nothing in METABRIC.

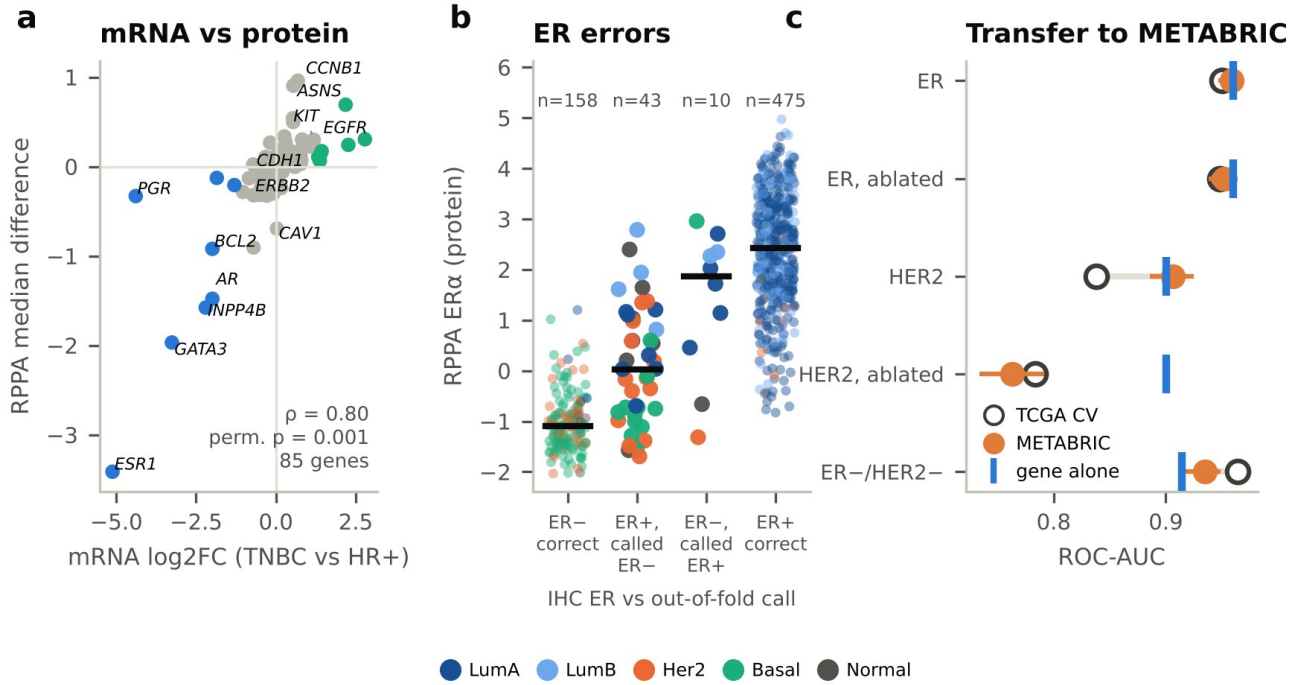


Figure 4. Validation. (a) DE log2FC vs RPPA median difference (TNBC vs HR+/HER2−); coloured, DEGs. (b) RPPA ERα by IHC ER status and out-of-fold call, coloured by PAM50. (c) AUC in TCGA cross-validation and in METABRIC for the frozen models, with the receptor gene alone in METABRIC.

4.5 Survival (beyond the talk)

In TCGA (median follow-up 2.4 years), PFI differed between the groups (log-rank p = 0.0072; Fig. 5a). After adjusting for age and stage, TNBC had a hazard ratio (HR) of 2.69 (95% CI 1.70–4.26) against HR+/HER2−, whereas HER2+ did not differ (HR 0.98, 95% CI 0.56–1.71; Fig. 5c). The group effect met the proportional-hazards assumption (p = 0.49), but age did not (p = 0.03). In METABRIC (1,846 tumours, 492 breast-cancer deaths within 10 years; Fig. 5b), HER2 gain carried an HR of 1.90 (95% CI 1.54–2.35). The ER−/HER2− effect violated proportional hazards. Split at 5 years, it was 2.46 in years 0–5 and 0.51 in years 5–10 (interaction p = 6×10^{-7}), which is the known early-relapse pattern of ER-negative disease. The ER−/HER2− group predicted by the frozen TCGA classifier also separated survival in METABRIC (log-rank p = 3×10^{-8}), and the METABRIC PAM50 subtypes followed the risk order reported by Nielsen et al. [25] (Table 2).

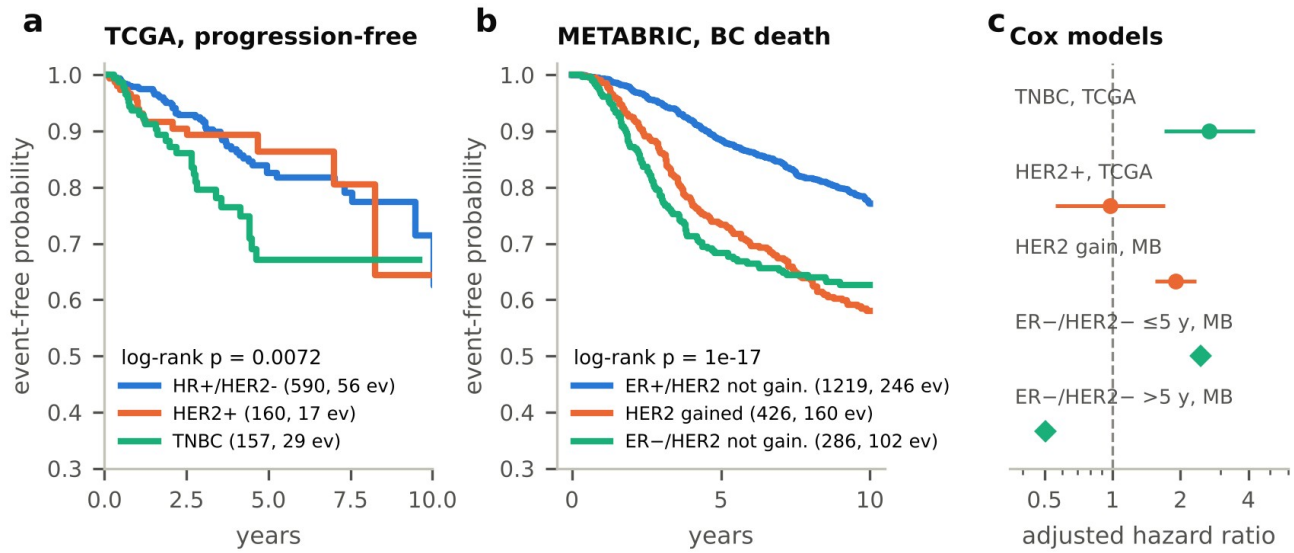


Figure 5. Outcome by group. Kaplan–Meier curves for (a) TCGA progression-free interval and (b) METABRIC breast-cancer death within 10 years (n, events). (c) Adjusted Cox hazard ratios (95% CI) against HR+/HER2−; diamonds, METABRIC ER−/HER2− effect split at 5 years. MB, METABRIC.

5. Conclusion and discussion

Summary. (1) The main structure in breast tumour expression is the ER axis. Methods that do not use labels recover TNBC as a basal-like group, but they do not recover HER2+ tumours as a group. (2) TNBC is a distinct transcriptional state: 2,741 genes differ more than 2-fold from HR+/HER2− tumours, and this list is stable to covariates, histology and label noise. (3) Whole-transcriptome classifiers perform as well as *ESR1* and *ERBB2* alone, but not better. (4) The results agree with protein data, with an independent cohort whose labels come from other assays, and with published numbers.

Biological interpretation. ER status behaves as a state of the whole regulon. Removing *ESR1* and every PAM50 gene leaves the prediction intact, because *GATA3*, *FOXA1*, *AGR3* and *XBP1* move together with it. In TNBC this programme is replaced by basal transcription factors (*FOXC1*, *EN1*, *SOX10*) and high proliferation and MYC activity, and these tumours relapse early, as the time-dependent hazard in METABRIC shows. HER2 positivity, on the other hand, is mostly a local genomic event. The amplicon dominates both the DE list and the classifier, and the rest of the transcriptome resembles HR+ disease. In practice, an expression assay adds little to IHC for ER and HER2 calls. Tumours in which mRNA and IHC disagree are still worth a second look, since our ER error analysis suggests that some of the IHC calls, not the model, are wrong.

Limitations. (i) TNBC and HR+ tumours differ in tumour purity and immune infiltration, which contribute to the proliferation and immune signals, and DE was not adjusted for purity. (ii) Pathway significance shrinks once inter-gene correlation is taken into account, so the pathway findings should be read as effect sizes rather than tests. (iii) The HER2 labels are noisy (some rest on status text alone), and the TCGA HER2 rule is our own. (iv) METABRIC has no PR data and uses copy-number HER2, so its groups are not identical to the clinical ones. (v) TCGA follow-up is short (median 2.4 years), and treatment was not modelled. (vi) The classifier comparisons rest on overlapping folds; the corrected test handles this conservatively.

Future directions. DE could be adjusted for purity (e.g. with ESTIMATE scores), or the immune signal could be deconvolved. It would be useful to test whether *ERBB2* mRNA can resolve equivocal IHC 2+ cases, and to extend the protein check to the CPTAC mass-spectrometry proteome. Finally, the HR+/HER2− group could be split into proliferative and immune subgroups and their outcome tested in METABRIC.

Data and code availability. TCGA data from UCSC Xena (GDC and TCGA hubs); METABRIC data from the cBioPortal datahub, used under the Open Database License (ODbL). Code, parameters, results and logs: https://github.com/Vijayavallabh/BT3041_Project. AI prompts and AI chat histories are shared privately with the course instructors as supplementary material. **Use of AI:** an AI assistant helped plan the analysis, write and run the code, review the results and draft

this report; every number was checked against the result files and every reference against PubMed.

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